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Smart Data | Green Agriculture | Sustainable Future



Using data and innovation to grow green vegetables, create jobs and improve nutrition in Rwanda

- Smart Farming
- Data Insights
- Job Creation
- Better Nutrition

Team Name
C&C_ECO INNOVATORS

University
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Green Today • Brighter Tomorrow

AGRI-HEALTH HUB RWANDA SNOPSIS



*AGRI-HEALTH HUB RWANDA*is a sustainable agricultural project that aims to grow and sell green vegetables, such as spinach,amalanth leaves(dodo),cassava leaves and carrot. to improve access to nutritious food, create jobs, and generate profit. We will transform biodegradable waste into organic manure to reduce pollution and maintain soil fertility.

Our farming system will use greenhouses to protect crops during rainy seasons and irrigation to supply water during sunny and dry periods.
"When we turn waste into wealth, we grow more than vegetables; we grow opportunities."

Through this approach,
AgriHealth Hub Rwanda seeks to promote better nutrition, responsible waste management, environmental protection, and sustainable economic growth through employment opportunities.

AgriHealth Hub Rwanda

Turning Agricultural Resources into Health, Income and Sustainable Solutions

AgriHealth Hub Rwanda is a data-driven digital platform designed to connect agriculture, nutrition, health, engineering, finance, business, and environmental sustainability to support community development in Rwanda.

The platform uses agricultural, nutrition, environmental, climate, population, and market data from the National Institute of Statistics of Rwanda (NISR) and other reliable public sources to identify agricultural opportunities, nutrition-risk indicators, and areas requiring targeted interventions.

Agriculture and Nutrition: AgriHealth Hub recommends locally appropriate, nutritious crops based on available evidence, agricultural conditions, and community needs. It promotes food and nutrition security by helping communities understand the importance of dietary diversity and nutritious foods.

Farmer Training and Productivity: The platform provides practical agricultural training on land preparation, planting, soil fertility, water management, pest and disease control, climate-smart farming, harvesting, and storage. It also supports production tracking to assess training participation, farming practices, and changes in agricultural productivity.

Waste-to-Value Innovation: AgriHealth Hub identifies potential uses for agricultural residues, including composting, suitable animal-feed applications, and bioenergy. Recommendations depend on technical feasibility, safety, and local conditions. The goal is to reduce waste and turn agricultural resources into valuable products.

Engineering and Technology: The platform includes a proposed smart processing module that could eventually use sensors to monitor temperature, moisture, and processing conditions. Its initial prototype can demonstrate these functions using clearly labeled simulated data.

Business and Finance: A business calculator estimates processing costs, operating expenses, potential revenue, and profit margins. Business dashboards help users monitor collection, processing, storage, distribution, and sales while exploring opportunities for income generation and employment.

Health and Environmental Impact: The platform promotes responsible waste management, cleaner surroundings, safer handling practices, and improved access to

nutritious foods. It aims to support better food and nutrition security without replacing professional medical diagnosis or treatment.

By connecting farmers, households, cooperatives, businesses, researchers, and potential partners, AgriHealth Hub Rwanda seeks to transform agricultural challenges into sustainable opportunities. Its long-term vision is to use reliable data, practical training, appropriate technology, and responsible business models to improve agricultural productivity, strengthen livelihoods, reduce waste, and support healthier communities.

All prototype estimates and simulated indicators will be clearly identified until verified datasets and real-world results become available.

PROBLEM STATEMENTS

"When food is difficult to access, waste is poorly managed, and employment opportunities are limited, both people and the environment are affected."

Rwanda continues to face challenges related to nutrition, unemployment, waste management, soil conservation, and sustainable vegetable production. According to Rwanda's 2024 Comprehensive Food Security and Vulnerability Analysis, 30% of children under five were stunted, highlighting the continuing need to improve nutrition and access to diverse foods. NISR's

Labour Force Survey reported a youth unemployment rate of 18.5% in 2024, reflecting the importance of creating income-generating opportunities. Furthermore, NISR's 2023–2024 household survey found that 38.5% of households disposed of rubbish in fields or bushes, while biodegradable waste that could be composted may be lost as a valuable agricultural resource. Soil protection and water availability are also important challenges for farmers. NISR's Season A 2025 agricultural survey reported that only 13.4% of farmers practiced irrigation, highlighting the importance of appropriate water-management practices.

Project Goal and Objectives

4.1 Overall goal

To improve sustainable agricultural productivity, access to nutritious vegetables, organic waste recovery and income-generating opportunities in Rwanda through integrated farming, appropriate technology and responsible business practices.

4.2 Specific objectives

1. Establish a pilot farm producing locally appropriate nutritious vegetables.
2. Introduce suitable irrigation, soil conservation and climate-smart agricultural practices.
3. Collect and convert approved biodegradable materials into safe, mature compost.
4. Train participating farmers and young people in agricultural production, waste management and entrepreneurship.

5. Develop a digital platform for production monitoring, market information and business calculations.
6. Create direct and indirect employment opportunities through farming, processing, packaging, transport and marketing.
7. Promote nutrition awareness and safe food handling in participating communities.
8. Establish a measurable business model that can be improved and expanded after the pilot phase.

Project Components and Activities

Component 1: Sustainable vegetable production





The project will initially focus on four crops:

- Spinach.
- Amaranth leaves (dodo).
- Cassava leaves.
- Carrots.

Activities will include land preparation, seed selection, planting, crop management, soil improvement, pest and disease monitoring, harvesting, sorting, packaging and marketing.The crops will be selected according to soil

suitability, climate, expected production costs, nutritional value, customer demand and available water.

Cassava leaves will be harvested and prepared according to safe food-handling practices, including appropriate cooking; they will not be marketed as ready-to-eat raw leaves.

Component 2: Greenhouse and irrigation system





A small greenhouse may be established to protect selected crops from some adverse weather conditions and support better production management. The project will also evaluate drip irrigation, safe rainwater harvesting, mulching and soil-moisture monitoring.

Activities:

1. Assess water availability and soil conditions.

2. Design a suitable irrigation layout.
3. Install storage tanks and irrigation equipment as required.
4. Establish a greenhouse only after comparing its cost with expected benefits.
5. Train operators in equipment maintenance and water-efficient farming.
6. Record water use and crop production to evaluate performance.

Component 3: Organic waste recovery and composting









The project will develop a small composting unit for approved biodegradable materials.

Activities:

- Identify reliable sources of clean vegetable residues and other suitable organic materials.
- Agree on collection schedules with suppliers.
- Separate organic materials from plastics, metals, glass and hazardous waste.
- Prepare compost mixtures and manage moisture, aeration and temperature.
- Monitor maturity and quality.
- Use finished compost on the farm and explore sales to other farmers.

Compost will not be sold as safe or mature until the necessary quality checks have been completed. Where required, laboratory testing and applicable product regulations will be followed.

Component 4: Digital Agri-Health Hub platform

The digital platform will connect farming, environmental management, training and business monitoring.

Farmer learning centre

Practical lessons, crop calendars, irrigation guidance, soil fertility, pest management and harvesting techniques.

Farm monitoring

Records of land area, seeds, labour, water use, harvest volumes, sales and production costs.

Market information

Customer orders, available produce, verified market prices and opportunities to connect farmers with buyers.

Business calculator

Revenue, costs, gross margin, net profit, compost production costs and cash-flow monitoring.

Smart monitoring prototype

Demonstration of temperature and moisture monitoring using simulated readings at first, with real sensors introduced after testing and funding.

The initial prototype will prioritise functions that are inexpensive, useful and easy to test. The project will not claim that simulated readings represent real farm measurements.

Component 5: Nutrition and community awareness

The project will organise awareness sessions on dietary diversity, the value of vegetables, safe food preparation and proper food storage.

It will seek technical input from qualified nutrition and health professionals. It will not diagnose malnutrition or replace professional medical care.

Component 6: Business development and farmer training

Training will cover:

- Farm planning and record-keeping.
- Crop production and harvesting.

- Composting and safe waste handling.
- Product packaging and marketing.
- Basic bookkeeping and financial literacy.
- Customer service and cooperative marketing.
- Digital platform use.

Training may be delivered directly by the project or through qualified partners, depending on available resources.

6. Target Beneficiaries

The initial project will prioritise the following groups:

Beneficiary group	Expected benefit
Smallholder farmers	Training, improved practices and potential market connections
Youth and women	Opportunities for paid work and small-business development
Households	Improved opportunities to access fresh vegetables
Farmer cooperatives	Compost supply, technical training and coordinated sales
Restaurants and retailers	Potentially reliable local vegetable supply

Schools and institutions	Potential suppliers and customers, subject to agreements
Environment and surrounding communities	Better recovery of suitable organic waste and improved resource management

Actual beneficiary numbers will be established after the project site and operating capacity are confirmed.

7. Implementation Plan

The project will use a 12-month pilot schedule.

Period	Main activities	Expected output
Months 1–2	Site selection, market research, soil and water assessment, partnerships and budgeting	Confirmed pilot design and implementation plan
Months 2–3	Land preparation, irrigation setup, initial planting and composting-area preparation	Operational farm and initial waste-recovery system
Months 3–4	Digital prototype development and initial farmer training	Basic platform and training records
Months 4–6	Crop management, compost processing, first sales when crops are ready	Initial production and sales records
Months 7–9	Improve production scheduling, customer relationships and compost quality	More reliable operating processes

Months 10–12	Financial review, environmental monitoring and project evaluation	Pilot evaluation and scale-up decision

The schedule will be adjusted to crop maturity, weather conditions, funding availability and the time needed for compost to mature.

8. Management and Staffing

The proposed pilot team will consist of the following roles. Some roles may initially be part-time or combined to reduce operating costs.

Role	Main responsibility
Project manager	Overall leadership, partnerships, reporting and coordination
Agricultural officer	Crop planning, farm operations and technical guidance
Farm workers	Planting, irrigation, crop care and harvesting

Composting operator	Waste sorting, compost processing and quality records
Digital developer or technician	Prototype development, maintenance and data management
Sales and marketing officer	Customer relations, sales and distribution
Finance and administration officer	Bookkeeping, expenditure control and financial reports
Nutrition or environmental adviser	Technical guidance through suitable partnerships

The project will comply with applicable labour, occupational safety and business requirements.

9. Business Model and Revenue Generation

Agri-Health Hub Rwanda will use a mixed-revenue model so that the project does not depend entirely on donor funding.

9.1. Revenue streams



Vegetable sales

Sale of fresh produce to households, retailers, restaurants and institutional customers.



Compost sales

Sale of quality-controlled compost to farmers and cooperatives after the product meets applicable requirements.



Training services

Paid training contracts with organisations, cooperatives and institutions that can finance the service.



Digital services

Future subscriptions, business reporting or farm-management services, introduced only after demand has been validated.

9.2. Marketing strategy

The project will use a direct and partnership-based marketing approach.

1. Identify potential customers before planting large quantities.
2. Interview vegetable retailers, restaurants, households and institutions about their preferred products, quantities, prices and delivery schedules.
3. Provide samples and product information where appropriate.
4. Develop a simple brand identity and packaging that preserves freshness.
5. Use WhatsApp, social media and the digital platform to receive orders.
6. Track customer satisfaction, repeat purchases, rejected produce and payment delays.
7. Establish written supply arrangements where feasible.

The business will avoid producing large volumes solely on the assumption that customers will buy them.

10. Market Analysis

The project will investigate four market segments:

Market segment	Customer need	Proposed value
Households	Fresh, convenient vegetables	Local produce and simple nutrition information

Restaurants and food vendors	Regular deliveries and predictable quality	Planned supply and customer communication
Retailers and market traders	Products they can resell	Graded vegetables and coordinated delivery
Farmers and cooperatives	Affordable soil-improvement inputs	Quality-controlled compost and training

Before launch, the team will carry out a small market survey to establish:

- Which vegetables are in greatest demand.
- Average selling prices for different grades and seasons.
- How much customers purchase per week.
- Competitors and their selling methods.
- Transportation, storage and packaging costs.
- Potential demand for compost.
- Willingness to pay for training and digital services.

The survey findings will determine the crop mix, production volume and revenue assumptions in the final business plan.

11. Preliminary Budget for the 12-Month Pilot

The following is an illustrative planning budget in Rwandan francs (RWF) for a small pilot. It assumes access to a modest site, a small greenhouse, a basic

irrigation system, a limited composting operation and a simple digital prototype.

These amounts are preliminary allowances for planning, not verified supplier quotations. Land purchase, major construction, vehicle purchase and a fully developed commercial software platform are excluded.

No.	Budget item	Estimated cost (RWF)
1	Site preparation and land-use allowance	1,200,000
2	Small greenhouse and installation	4,500,000
3	Irrigation system, storage and fittings	1,500,000
4	Seeds, seedlings and initial farm inputs	700,000
5	Composting area and basic equipment	1,000,000

6	Farm tools and protective equipment	800,000
7	Harvesting, storage and packaging equipment	1,200,000
8	Digital prototype and basic dashboard	2,500,000
9	Training and community engagement	1,000,000
10	Labour and operational support	3,600,000
11	Transport and distribution	1,000,000
12	Monitoring, testing and evaluation	500,000
	Subtotal	19,500,000
	Contingency allowance (approximately 8%)	1,560,000

Indicative total pilot budget	21,060,000 RWF

Preliminary funding requirement

21,060,000 RWF

About 21.06 million RWF for the illustrative pilot budget. The final request must be recalculated using quotations, site costs, available contributions and the funder's eligible expenses.

11.1. Proposed use of funds

- Agricultural production: land preparation, greenhouse, irrigation, seeds and farm tools.
- Waste recovery: composting infrastructure, equipment and quality checks.
- Digital innovation: basic software development, data management and testing.
- Community and workforce development: training, labour and awareness activities.
- Distribution and accountability: transport, packaging, monitoring and contingency.

The project should obtain quotations from at least two or three suppliers for major items before submitting the final budget.

12. Financial Projections and Profitability

The pilot will need to distinguish revenue from profit. Money received from vegetable sales is not the same as money available after paying production, labour, transport, packaging, rent, maintenance and other costs.

12.1. Revenue calculation

The main formula is:

Revenue=Quantity sold * Selling price

For example, if the farm sells 500 kg of vegetables at an illustrative average price of 1,000 RWF per kilogram:

$$500 \times 1,000 = 500,000 RWF$$

This is revenue for the stated quantity, not necessarily monthly revenue and not profit.

12.2. Profit calculation

Gross profit= Sales revenue-Direct costs

Net profit= Total revenue-All operating costs

Revenue from compost and training should be included only when sales or paid contracts are actually obtained.

12.3. Financial controls

The project will maintain:

- Daily records of sales and expenses.

- Production records for each crop.
- Separate records for compost production and sales.
- Receipts and invoices for purchases.
- Monthly cash-flow statements.
- Stock records for seeds, packaging and other inputs.
- A monthly comparison of actual results against the budget.

12.4. Break-even analysis

The project will calculate the sales volume required to cover operating costs.

$$Break - evenquantity = \frac{Fixedcosts}{Sellingpriceperunit - Variablecostperunit}$$

This calculation will be completed after real production costs and market prices are available. The project should not assume that the full initial investment will be recovered during the first year.

13. Monitoring, Evaluation and Key Performance Indicators

The project will use a monitoring and evaluation framework to track its performance.

The following are proposed pilot targets, to be confirmed after the baseline assessment and available resources are known.

Indicator	Proposed 12-month target	Means of verification

Pilot farm established	1 operational pilot farm	Site inspection and farm records
Crops introduced	4 planned vegetable crops, subject to suitability	Planting and harvest records
Farmers trained	At least 30 participants	Attendance and training records
Digital prototype	1 functioning basic prototype	Demonstration and test records
Organic waste recovery	Measured and recorded throughout the pilot	Weighing and collection logs
Compost production	At least 3 documented batches, if inputs and conditions allow	Batch records and quality checks
Direct employment	Target of 3–5 paid roles, subject to funding	Contracts and payroll records

Customers	At least 10 repeat buyers or purchasing accounts sought	Sales ledger and customer records
Financial performance	Monthly revenue, costs and margins documented	Financial statements
Environmental practices	Water use and waste recovery monitored	Farm and composting logs

These targets are planning goals, not achieved results or guaranteed outcomes.

Reporting schedule

- Weekly: crop production, harvest and waste collection records.
- Monthly: financial performance, sales, expenses and operational challenges.
- Quarterly: farmer participation, customer feedback and environmental indicators.
- At month 12: final pilot evaluation and recommendation on whether to expand, revise or discontinue particular activities.

14. Environmental and Social Safeguards

The project will apply safeguards to ensure that its environmental and social goals do not create new risks.

Environmental safeguards

- Accept only suitable and traceable biodegradable materials for composting.
- Prevent leachate and contaminated runoff from reaching waterways.
- Use water efficiently and avoid unnecessary groundwater extraction.
- Store farm chemicals safely and follow approved handling instructions.
- Apply soil-conservation practices and minimise avoidable waste.

Food safety safeguards

- Use clean harvesting and packaging equipment.
- Protect harvested vegetables from contamination.
- Use safe water for washing produce where washing is required.
- Follow appropriate food-handling and storage practices.
- Seek relevant approvals and comply with applicable standards.

Social safeguards

- Provide safe working conditions and protective equipment.
- Promote fair access to training and employment opportunities.
- Maintain clear financial records and transparent procurement.
- Protect personal data collected through the digital platform.
- Ensure that nutrition messages are evidence-based and do not replace professional healthcare.

15. Risk Analysis and Mitigation

Risk	Potential effect	Mitigation

Drought or excessive rainfall	Lower crop yields	Efficient irrigation, crop planning and suitable greenhouse protection
Crop pests and diseases	Production losses	Regular monitoring and integrated pest management
Market price fluctuations	Reduced revenue	Diversified customers, staggered planting and careful sales planning
High start-up costs	Delayed implementation	Begin with a small pilot and prioritise essential equipment
Contaminated waste	Unsafe compost	Source screening, separation and appropriate quality controls
Limited digital adoption	Low platform use	Simple design, practical training and offline record options
Delayed payments	Cash-flow problems	Written sales terms, credit limits and cash-flow monitoring

Insufficient funding	Incomplete activities	Phased implementation and diversified funding sources

16. Sustainability and Expansion Strategy

The project will pursue three dimensions of sustainability.

Financial sustainability: Income from produce, compost and appropriate services will gradually contribute to operating costs. Spending will be reviewed against actual sales performance.

Environmental sustainability: The project will promote organic resource recovery, soil improvement, water efficiency and responsible handling of agricultural residues.

Social sustainability: Training, local employment, partnerships and inclusive participation will help distribute the benefits of the project.

Expansion will take place only when the pilot demonstrates reliable production, sufficient demand, acceptable product quality and a credible financial model. Future stages may introduce additional farmers, more composting capacity, real sensors and a wider digital marketplace.

17. Funding Strategy and Partnership Plan

Agri-Health Hub Rwanda will seek a combination of grant funding, technical partnerships, community contributions and business revenue.

Potential funding and partnership channels include:

- Agricultural and environmental grant programmes.
- Youth entrepreneurship and innovation competitions.
- Agricultural cooperatives and farmer organisations.
- Private-sector partners in food distribution and waste recovery.
- Universities and research organisations.
- Financial institutions and responsible investors.
- Local development programmes and relevant public institutions.

The project will approach potential funders with a clear proposal, a verified budget, evidence of customer demand, an implementation schedule and measurable indicators.

No funding or partnership should be presented as confirmed until a formal agreement has been obtained.

18. Expected Results and Long-Term Impact

The project expects to contribute to the following outcomes:

Improved agricultural productivity

Better production planning, appropriate irrigation and improved soil management.

Reduced organic waste

Suitable biodegradable materials recovered and converted into useful compost.

Employment and entrepreneurship

Paid roles and business opportunities in farming, processing, distribution and technology.

Nutrition awareness

Improved community understanding of dietary diversity and safe vegetable preparation.

Digital agricultural innovation

Better records, market information and financial decision-making.

These outcomes will be evaluated using verified project data. The project will not claim that it has reduced national malnutrition, unemployment or pollution based solely on the establishment of a pilot farm.

19. Conclusion

Agri-Health Hub Rwanda proposes an integrated approach to sustainable agriculture, nutrition awareness, organic waste recovery, environmental protection, employment and digital innovation.

By combining vegetable production, composting, efficient water use, farmer training, market connections and digital business tools, the project aims to turn agricultural challenges into practical economic and environmental opportunities.

The proposed 12-month pilot will provide an opportunity to test the model, establish a reliable customer base, measure costs and revenue, evaluate compost quality and document the project's social and environmental contribution.

Its long-term success will depend on sound financial management, safe production practices, reliable data, community participation and strong partnerships.

Agri-Health Hub Rwanda aims to demonstrate that agricultural resources can create more than food: they can create jobs, protect the environment and support healthier communities.

Turning Agricultural Resources into Health, Income and Sustainable Solutions.

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